CPCS-203 MID 202

Q1/ multiple choices

1.

```
String s1 =new String("Hi");
String s2= new String("Bye");
System.out.println(s1.compareTo(s2));
```

H =

B = 66

72 − 66 = 6

A) false B) negative number C) 0 D) positive number

2.

```
String s1 = new String("Chat Gpt");
String s2 = "Chat Gpt";
System.out.print(s1 == s2);
System.out.println(" " + s1. equals(s2));
```

A) false false B) true true C) true false D) false true

3. `String c1 ="C,L;O".replaceAll(",;" , "#" ) + " ");`

A) C L O B) C#L#O C) C,L;O D) C,#L;#O

4. What method should we use to read full directory of a file?

A) getName() B) getPath() C) getAbsolutePath() D) getCanonicalPath()

5. Rabbit [ ] rabbit = new Rabbit[4] what is the right way to invoke the method eat() by this array of Rabbit ?

A) Rabbit.eat() B) rabbit.eat() C) rabbit[0].eat() D) this.eat()

the class

this is array

Inside the class

6. The output for System.out.println("c:\\Users\\pc\\eclipseworkspace\\arna\\program.txt".getName());

a) c:\\Users\\program.txt
b) c:\\Users
c) c:\\program.txt
d) program.txt

7. How to get the length of one dimensional array called student?

2D Array

a) System.out.println(“the length is: “ + student[i].length);

b) System.out.println(“the length is: “ + student.length);

c) System.out.println(“the length is: “ + student[i].length());

d) System.out.println(“the length is: “ + student.length());

not for Arrays

8. chose the correct output:

0 1 2 3 4

```
String [] s = "abc,abc?abc.abc abc#".split("[, ?.]");

          for(int i=0; i<s.length; i++)
               System.out.print(s[i] + " ");
```

A) abc abc abc abc abc

B) abc,abc abc.abcabc#

C) abc abc abc abcabc

D) abc abc abc abc abc#

9. `StringBuilder x = new StringBuilder ("Hello World");`
`System.out.println(x.insert(5,"HTML and Java"));`

`A) HelloHTML and Java World`

`B) HellHTML and Java o World`

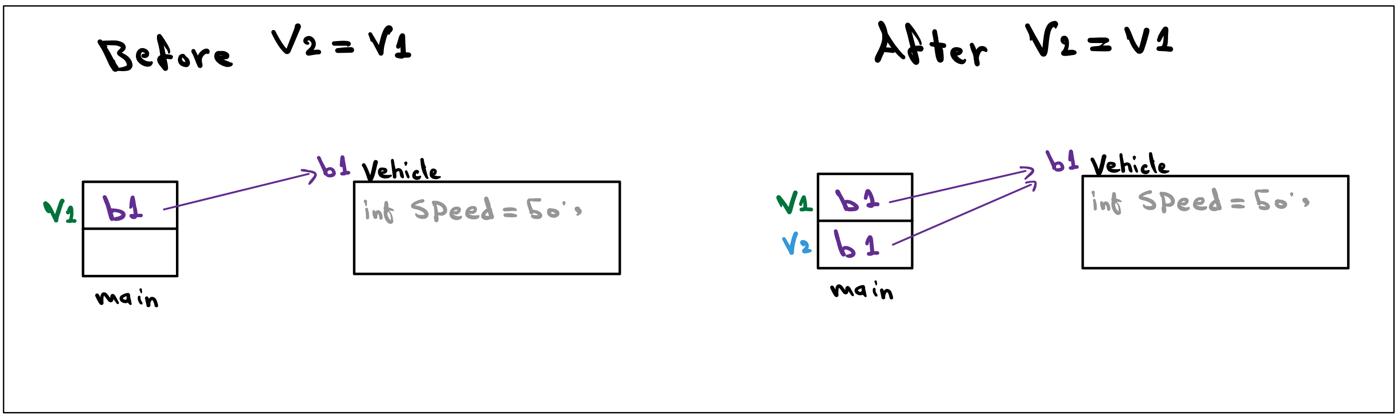
`C) HelloHTML and Java World`

`D) HelloHTML and JavaWorld`

Q2/ output

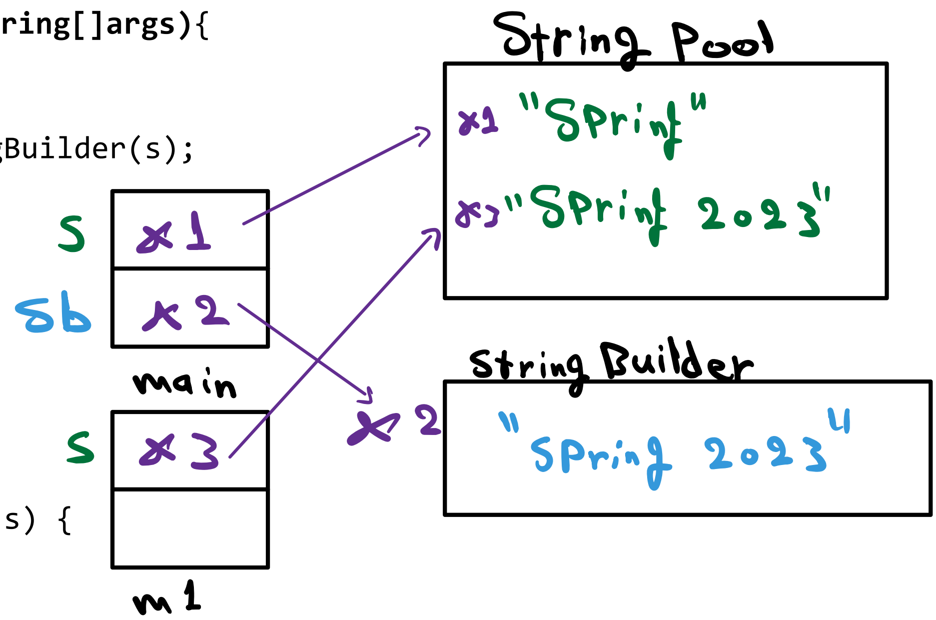
1. explain what happened in writing <u>or</u> drawing:

1 Vehicle V1 =new Vehicle(50);
Vehicle V2=V1;



2. Write the output:

1.
```
public static void main( String[]args){
String s = "Spring";
StringBuilder sb = new StringBuilder(s);
m1(s);
sb.append(" 2023");
System.out.println(s);
System.out.println(sb);
}
public static void m1(String s) {
s = s + " 2023";
}
```



Spring
Spring 2023

2.
```
public class A{
private String name;
private boolean attendance;
private int ID;
public static void main(String[]args){
A a=new A();
System.out.println(a.name);
System.out.println(a. attendance);
System.out.println(a.ID);
}
}//end of class A
```

null
false
0

1. Create a file called midTerm.txt and write the details as shown in the figure. (The file doesn't exist):

midTerm.txt
Amal Abdullah 20

```java
import java.io.*;
```

```java
public class Mid23 {

    public static void main(String[] args) throws IOException {
        File f = new File("midTerm.txt");
        PrintWriter pen = new PrintWriter(f);
        pen.append("Amal Abdullah 20");
        pen.flush();
        pen.close();
    }
}
```

2. Read from the previous file (text.txt) and print the details first name, last name, score. Then show the output on the console (The file exist):

```java
import java.io.*;
import java.util.Scanner;
```

```java
public class Mid23 {

    public static void main(String[] args) throws IOException {
        File f = new File("text.txt");
        Scanner read = new Scanner(f);
        String fName = read.next();
        String lName = read.next();
        int score = read.nextInt();
        System.out.println("First name: " + fName +
                "Last name: " + lName +
                "Score: " + score);

    }
}
```

Output:

```
First name: Amal Last name: Abdullah Score: 20
```

Q4/ Complete the following class according to the instructions, and at the end answer the question about whether the class is immutable or mutable:

```
public class Employee {
      private String name;
      private int id;
      private String department;
      private static int count; //to count employees' number

   public Employee() {
      //complete the method
      //invoke 2 arguments constructor
      //note that the default values for name and id are "unknown" and 0

        this("unknown",0);
      }

      public Employee (String name, int id) {
            //complete the method
            //invoke full arguments constructor
            //note that department default value is "general"
            this(name,id,"general");
      }

      public Employee (String name, int id, String department) {
            //Initialize data fields
            this.name=name;
            this.id=id;
            this.department=department;
            count++;
      }

      public String getName() {
            return name;
      }

      public void setName(String name) {
             this.name=name;
      }

      public int getId() {
            return id;
      }

      public void setId(int id) {
            this.id=id;
      }
      public static int getCount() {
            return count;
      }
      //complete the method
      public String info() {
            return "Name: "+getName()+" id: "+getId()+" department: "+
                      this.department +" count: "+getCount();
      }
}
```

Question: is the previous class mutable or immutable? And explain why:

No, it's not immutable because there is a setter method, which allows us to change the value after initializing it. To make it immutable, we need to remove all setter methods. This way, we won't be able to change the value after initializing it in the constructor.